[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SANDER, DISK, PNEUMATIC, PORTABLE (RANDOM ORBITAL)

This Commercial Item Description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. Scope. This commercial item description covers portable, pneumatic, random orbital disk sanders used for blending and smoothing filler, shaping, forming and feather edging on wood, metal, and plastic surfaces. The sander's random orbital action prevents scratches and swirl marks.

2. Salient Characteristics.

2.1 Design. The sander shall be similar to Figure 1 and comply with the performance criteria in Table I. The sander shall be of the long-handle type as depicted in Figure 1. The sander shall be equipped with a 6 inch diameter neoprene sponge pad with a canvas suitable for use with glue-on abrasive discs. The sander shall operate properly at the manufacturer's recommended operating pressure.

Table I

National Stock Number	Speed at No Load RPM	Air Consumption at No Load CFM (max.)	Horsepower (min.)	Pad Size Diameter inch (nominal)
5130-00-204-0623	8,000 - 12,000	20	0.14	6

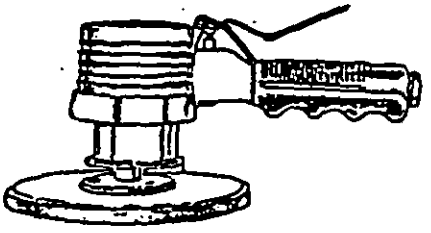


Figure 1. Sander.

2.2 Materials: The materials used in the construction of all components shall be capable of completing the examination and tests required herein. All components shall be new and unused. Repaired or reconditioned components shall not be acceptable. A metal or composite housing is required. Cadmium is not an acceptable finish.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5130

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2.3 Motor: The sander shall be equipped with a rotary vane-type motor.

2.4 Throttle: The sander shall be equipped with a manually operated, lever-type throttle that shall automatically close the air inlet valve when actuating hand pressure is removed.

2.5 Lubrication: Provisions shall be made for lubrication to be supplied to the rotor, cylinder, and other moving parts in the path of the air stream. The oil may be applied manually into the motor or supplied automatically through the air supply line. Other suitable means shall be provided for supplying lubrication to gears and other moving parts not in the air stream.

2.6 Air inlet connection: The sander shall have a female, 1/4-inch size, NPT or NPTF air inlet connection. The inlet shall be equipped with an air filter of 20 mesh or finer which shall be effective in preventing solid particles in the compressed air supply from entering the tool. The filter shall be easy to remove, clean, and replace.

2.7 Bearings: The rotor shaft shall be provided with ball bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. If open type bearings are used, they shall be suitably housed and adequately sealed to prevent leakage of lubricant and any entrance of dirt and dust. Sleeve bearings are not acceptable.

2.8 Backup pad: Sanders shall be equipped with a pad for backing the abrasive disk of the size specified in Table I. The pad shall have a minimum thickness of 1/4 inch. It shall be made of a neoprene sponge and shall be capable of withstanding shock without permanent deformation or rupture. The pad shall have a canvas suitable for use with glue-on abrasive disks.

2.9 Random orbital motion: Sanders shall contain a counterbalanced offset drive spindle which shall produce a combined rotary and oscillating (random orbital) pad surface motion when the tool is operated under load.

2.10 Manual: One copy of a technical manual consisting of the manufacturer's standard commercial instruction and parts list shall be furnished with each tool.

2.11 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

3. Regulatory Requirements: In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

3.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

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4. Quality assurance provisions.

4.1 Responsibility for inspection: Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services adhere to prescribed requirements.

4.2 Inspection: It shall be determined that the sample units meet all salient characteristics stated in the requirements section of this document. Any sample containing one or more defects shall be rejected, and if the number of rejected units exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Test conditions: All tests shall be conducted with an air supply of 90 psig (620 kPa) or at the manufacturers recommended pressure measured as close to the tool as practicable. The test tool shall be connected to the air supply line by means of a 10 foot length of air hose having a hose size of 5/16 inch (inner diameter). The air supply line shall be equipped with a filter, air pressure regulator and lubricator.

4.4 First article inspection. When specified, the contractor shall furnish one sample of a completely fabricated tool for first article inspection. The sample shall have been manufactured and inspected using the same materials, processes, and methods that will be used in the production of the tools used to fulfill the terms of the contract or purchase order. The sample shall meet the requirements of this Commercial Item Description. Approval of the first article sample shall not relieve the contractor of the responsibility to furnish tools in accordance with the requirements of this Commercial Item Description. First article inspection shall consist of the following evaluation in the sequence listed: Inspection, Air consumption test, Speed test, Sound pressure level test, Practical service test, Horsepower test, Service test and Examination after service test.

4.5 Production lot inspection and testing: Production lot testing shall consist of the following evaluations conducted in the sequence listed: Inspection, Air consumption test, Speed test, and Practical service test. A sample lot shall consist of all items of same NSN offered for delivery at one time. Sampling procedures shall be in accordance with ANSI/ASQC Z1.4 at an Inspection Level of S-3. The acceptable quality level (AQL) shall be 1.5 defects per 100 units tested. A sample unit shall consist of one completely fabricated tool.

4.6 Air consumption test: Air consumption shall be measured with either a certified liquid displacement meter or a flowmeter.

4.7 Speed test: Each sample tool shall be tested to determine compliance with the speed requirement specified herein. A stroboscope, tachometer or other suitable device, of sufficient capacity and accuracy, shall be used.

4.8 Sound pressure level test: The Sound pressure level test shall be conducted in accordance with the applicable provisions of the Compressed Air and Gas Institute's, "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment". The sound pressure level shall not exceed 85 decibels on the A weighted scale.

4.9 Practical service test: Each sample tool shall be operated for 5 minutes. The test shall duplicate a typical work application. During the test, the tool shall be checked for throttle operation, rotation, excessive noise or vibration, ease of operation, and general serviceability.

4.10 Horsepower test: Each sample tool shall be tested to determine compliance with the horsepower requirement specified herein. The test shall be conducted on a prony brake analyzer or similar system of appropriate capacity calibrated to assure accurate readings. The tool shall be disassembled to the extend necessary to measure the torque and speed of the motor. Tool motors not developing the horsepower required in Table I shall be rejected.

4.11 Service test: A simulated service test shall be conducted for 150 hours. The test sample shall be installed in a fixture in a manner that the tool can be loaded and automatically actuated for 150 hours. A load of one pound, in addition to the

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weight of the tool, shall be applied perpendicular to the plane of the sanding pad during the load portion of the test cycle. Each cycle shall be 2 minutes ON and 2 minutes OFF.

4.12 Examination after service test: Tools satisfactorily completing the Service test shall be subjected to the Sound pressure level test, Air consumption test and shall conform to the applicable requirements of these tests. Upon completion of these tests, the tool shall be disassembled and the parts examined. Any tool having broken, cracked, or deformed parts shall be rejected.

5. Preservation, packaging, packing, and marking: Tools shall be preserved, packaged, packed, and marked as specified in the contract or purchase order.

5.1 Identification markings: Each item (including all items within a set or kit) shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

6. Notes:

6.1 The issue of referenced documents in effect on the date of invitation for bids or request for proposal shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the Contracting Officer.

6.2 Military standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.

6.3 ANSI standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.4 The "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment", is available from the Compressed Air and Gas Institute, 1300 Sumner Avenue, Cleveland, OH 44115

Military Coordinating Activity

Army - GL

Preparing Activity

GSA - FSS

Custodians

Army - GL
Air Force - 99

Review Activities

Air Force - 35, 84

User Activity

Navy - MC

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